Milestone Report

Active Feature-Value Acquisition with Sampled Expected Utility for Cost-Sensitive Learning

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In many real-world settings, acquiring feature values carries a non-trivial cost, and collecting complete data for all instances is unnecessary or infeasible. Active Feature-Value Acquisition (AFA) addresses this by iteratively selecting which missing values to acquire in order to maximize model improvement per unit cost. This report describes our implementation of the SEU framework from Saar-Tsechansky et al. [1] and presents experimental results under two missingness settings.

1. Data and Experimental Setup

We use the UCI Breast Cancer dataset,[1] which contains 286 instances with 9 categorical features and a binary class label. Nine rows with original missing values were removed, leaving 277 instances (196 no-recurrence, 81 recurrence). To simulate AFA under Scenario 1 (known labels), we randomly mask **10%** or **20%** of feature entries. Each experiment uses an 80/20 train/validation split, and results are averaged over 20 random seeds with full candidate scoring (sample_fraction = 1.0).

2. Implementation Details

So far, we implemented: (1) a Categorical Naive Bayes classifier that handles missing entries via marginalisation during fitting and prediction, with Laplace smoothing $\alpha = 1$; (2) the SEU scoring function $\mathrm{E}(q_{i,j}) = \sum_k U(F_{i,j} = v_k)\, P(F_{i,j} = v_k \mid y_i)$; (3) two candidate sampling variants, SEU-US (uniform) and SEU-ES (uncertainty-weighted); (4) SEU-Accuracy, an ablation that replaces log-gain with classification accuracy as the utility signal; and (5) a uniform random baseline and a fully observed upper bound. Performance is evaluated by validation accuracy at fixed budget fractions and by the area under the accuracy-vs-cost curve (AUC).

3. Results

Tables 1 and 2 report validation accuracy at four budget checkpoints and mean AUC across 20 trials. Figure 1 shows learning curves during the first 50% of the acquisition budget; Figure 2 shows the full curves. Shaded regions denote ± 1 standard deviation.

We highlight three observations. First, SEU-Accuracy achieves the highest validation accuracy across both settings, with the advantage most pronounced in the first 50% of the budget. At 25% budget, SEU-Accuracy reaches 0.761 (10%) and 0.769 (20%), compared to 0.744 and 0.742 for Uniform. Second, SEU-US and SEU-ES perform comparably to Uniform throughout, and are identical to each other (AUC = 0.739 at 10%, 0.730 at 20%). Third, all methods converge to the same final accuracy (0.742) at full budget, and SEU-Accuracy and Uniform periodically exceed the fully observed ceiling of 0.750 during acquisition.

[1]M. Zwitter & M. Soklic, *Breast Cancer* [Dataset], UCI ML Repository, 1988. https://archive.ics.uci.edu/dataset/14/breast+cancer

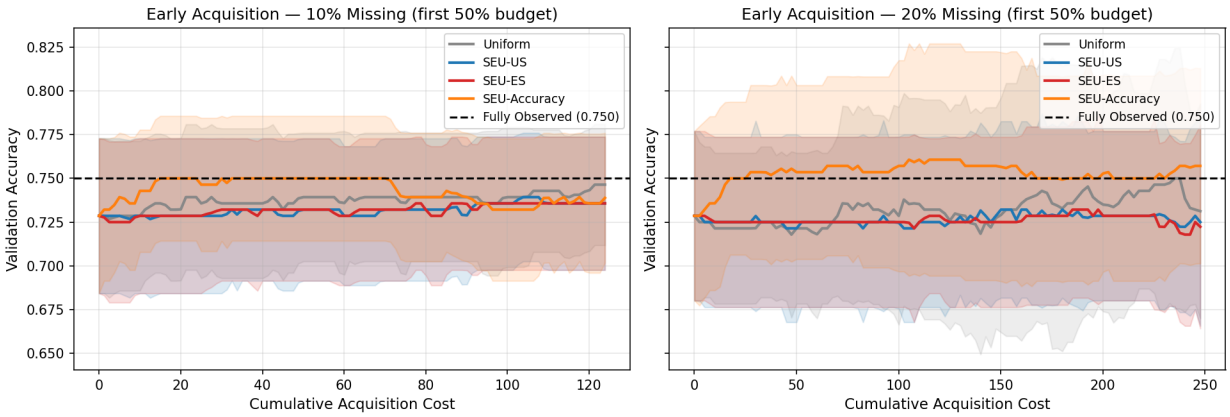

Figure 1 | Early-stage acquisition (first 50% of budget) under 10% (left) and 20% (right) missingness.

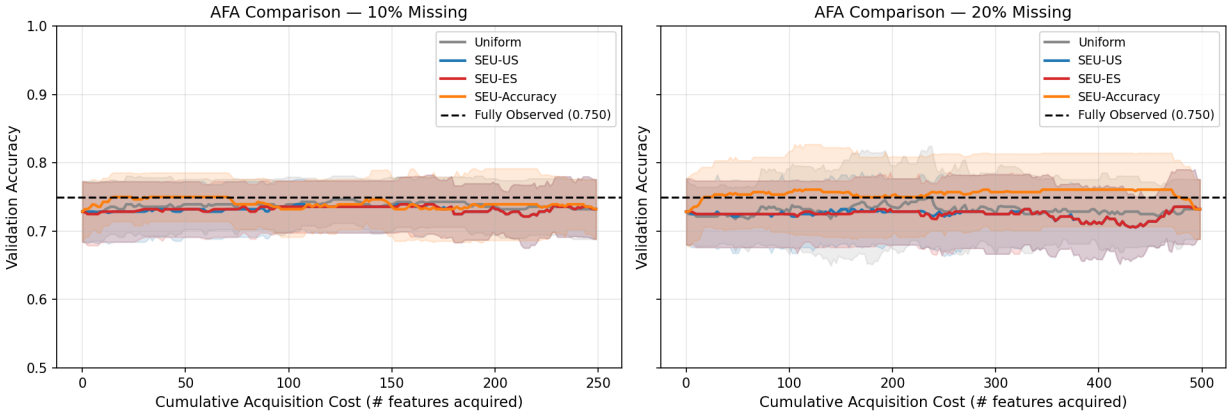

Figure 2 | Full acquisition curve under 10% (left) and 20% (right) missingness (∼500 rounds).

4. Discussion

Why does SEU-Accuracy outperform log-gain variants? The original paper predicts that log-gain, by capturing confidence changes that precede accuracy improvements, should provide a stronger utility signal than raw accuracy. Our results do not support this prediction. Diagnostic logs show that log-gain scores are near-zero or negative across all acquisition rounds. We think that this might occurs if the Naive Bayes model is already well-calibrated on the partially observed features: thus acquiring an additional feature redistributes posterior probability mass without meaningfully shifting the model's confidence in the correct class. Accuracy gain, as a discrete 0/1 signal, sidesteps this issue by directly identifying acquisitions that change the predicted label. Whether this finding generalizes beyond this dataset and classifier remains to be investigated.

Why do SEU-US and SEU-ES perform identically? SEU-US samples candidates uniformly, while SEU-ES prioritizes uncertain instances. In principle, SEU-ES should direct acquisition toward instances where the model is most confused, yielding faster improvement. In practice, both strategies score candidates using log-gain, which is near-zero or negative for all candidates. As a result, the max-scoring selection is effectively arbitrary in both cases, and the difference in sampling strategy has no impact on which feature is ultimately acquired.

Table 1 | Validation accuracy and AUC at 10% missingness (N=20 trials). Fully observed ceiling: 0.750.

Strategy	@25%	@50%	@75%	@100%	AUC
Uniform	0.744	0.746	0.744	0.742	0.742 ± 0.046
SEU-US	0.738	0.743	0.737	0.742	0.739 ± 0.046
SEU-ES	0.738	0.743	0.737	0.742	0.739 ± 0.046
SEU-Accuracy	0.761	0.761	0.759	0.742	0.758 ± 0.045

Table 2 | Validation accuracy and AUC at 20% missingness (N=20 trials). Fully observed ceiling: 0.750.

Strategy	@25%	@50%	@75%	@100%	AUC
Uniform	0.742	0.738	0.738	0.742	0.739 ± 0.047
SEU-US	0.732	0.728	0.731	0.742	0.730 ± 0.044
SEU-ES	0.732	0.728	0.731	0.742	0.730 ± 0.044
SEU-Accuracy	0.769	0.764	0.764	0.742	0.764 ± 0.051

Why do some methods exceed the fully observed upper bound? The fully observed upper bound represents the expected accuracy of Naive Bayes trained on complete data. Several methods periodically exceed this bound, which is counterintuitive. We attribute this to two factors. First, the dataset is small: with approximately 55 test instances, a single prediction flip changes accuracy by $\sim$1.8%, making the upper bound itself noisy across seeds. Second, missing entries may act as implicit regularization — reducing the effective feature count per instance can prevent Naive Bayes from overfitting to spurious feature correlations, occasionally yielding better generalization than the fully observed model.

5. Plan for Remaining Weeks

- **Investigate the log-gain pathology.** We will examine whether log-gain scores are systematically negative due to a structural property of Naive Bayes, or whether this is specific to this dataset. We will inspect per-round utility distributions and test whether the pattern reverses on a larger dataset with more heterogeneous features.
- **Adopt per-round error reduction as an additional metric.** Following the original paper, we will track error reduction at each acquisition round in addition to cumulative accuracy, which may better capture the marginal value of individual acquisitions.
- **Tune candidate pool size γ.** We will vary the fraction of candidates scored at each round and examine how pool size interacts with the choice of utility function, particularly whether a smaller pool degrades SEU-Accuracy more than the log-gain variants.

References

[1] Maytal Saar-Tsechansky, Prem Melville, and Foster Provost. "Active Feature-Value Acquisition". In: *Management Science* 55.4 (2009), pp. 664–684.